

# **Skills Passport**

## **A Human Capacity Intelligence System for Verifiable, Portable, and Opportunity-Linked Skills**

**Jamie Valenzuela-Mumau, EdD**

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### **Executive Summary**

Across education and workforce systems, there is increasing consensus that transferable human capacities—such as communication, collaboration, self-regulation, and problem solving—are critical predictors of long-term success. However, despite decades of research and widespread adoption of competency frameworks, these capacities remain inconsistently measured, rarely verified, and weakly connected to opportunity systems (National Research Council, 2012; World Economic Forum, 2023).

This paper argues that the limitation is not conceptual but structural. Existing systems lack the ability to translate human development into verifiable, developmentally grounded, and portable signals. As a result, learners are unable to demonstrate their capabilities in meaningful ways, and employers lack reliable information for decision-making.

In response, this paper proposes a Human Capacity Intelligence System, operationalized through the Skills Passport. The system integrates evidence-centered design, developmental progression models, multi-source verification, and interoperable credential infrastructure into a unified framework. This approach strengthens validity, improves fairness, and aligns learning with real-world opportunity structures (American Educational Research Association [AERA], American Psychological Association [APA], & National Council on Measurement in Education [NCME], 2014).

### **1. Background and Context**

Over the past two decades, research across learning sciences, psychology, and workforce development has converged on the importance of transferable human skills. The National Academies identified cognitive, interpersonal, and intrapersonal competencies as essential for life and work, emphasizing their role in adaptability and long-term success (National Research Council, 2012). Similarly, the World Economic Forum has highlighted skills such as analytical thinking, resilience, leadership, and collaboration as critical in an evolving global labor market (World Economic Forum, 2023).

Despite this convergence, systems of education and workforce signaling remain anchored in models designed for knowledge transmission rather than human capacity development. Traditional credentials, including grades and degrees, function as indirect proxies for capability but fail to capture how individuals apply skills in real-world contexts. This creates a persistent misalignment between what systems value, what they measure, and what they communicate to employers.

## **2. Problem Definition**

### **2.1 Measurement Misalignment**

Traditional assessment systems prioritize standardized, decontextualized measures of knowledge. While these approaches offer reliability and comparability, they are not well suited to capturing complex, context-dependent behaviors such as collaboration, initiative, or adaptive problem solving. Evidence-centered design research demonstrates that valid assessment requires alignment between claims, evidence, and tasks; however, most current systems rely on indirect proxies rather than observable performance (Mislevy, Almond, & Lukas, 2003; National Research Council, 2001).

### **2.2 Fragmentation of Evidence**

Human development occurs across multiple environments, including schools, workplaces, families, and communities. Ecological theories emphasize that development is shaped through interactions across these systems (Bronfenbrenner & Morris, 2006). However, existing systems capture evidence in isolation, preventing the formation of a coherent, longitudinal understanding of capability.

### **2.3 The Verification Gap**

The absence of verification mechanisms undermines trust in skills-based systems. Research demonstrates that observed performance and multi-source evidence are significantly more reliable indicators of capability than self-report (Miller, 1990; Pellegrino, Chudowsky, & Glaser, 2001). Without verification, systems cannot distinguish between demonstrated and perceived ability.

### **2.4 Absence of Developmental Representation**

Most systems treat skills as static attributes rather than developmental processes. However, competence develops progressively through increasing levels of independence, adaptability, and transfer (National Research Council, 2007; Zimmerman, 2002). Binary classifications obscure this progression and limit actionable insight.

### **2.5 System-Level Consequences**

These limitations result in inefficiencies across systems. Employers lack reliable hiring signals, learners cannot demonstrate growth effectively, and education systems struggle to align with workforce demands. This leads to underutilized human potential and inequitable access to opportunity.

### **3. Criteria for Effective Solutions**

A viable solution must be evidence-based (Mislevy et al., 2003), developmental (National Research Council, 2007), context-aware (Vygotsky, 1978), verifiable (Pellegrino et al., 2001), transparent (AERA et al., 2014), and portable across systems (World Wide Web Consortium [W3C], 2022; 1EdTech Consortium, 2023). No current system satisfies all of these criteria simultaneously.

### **4. Evaluation of Existing Approaches**

Existing approaches address portions of the problem but fall short of a comprehensive solution. Standardized testing offers reliability but lacks ecological validity. Academic transcripts provide structured records but are context-bound. Digital badges introduce portability but often lack verification rigor. Social-emotional frameworks provide conceptual clarity but are not operationalized into scalable systems.

### **5. Proposed Solution: Human Capacity Intelligence System**

The Skills Passport introduces a layered architecture grounded in evidence-centered design. Rather than collapsing interpretation into a single score, the system separates the inferential chain:

**Evidence → Context → Development → Capacity → Confidence**

This structure ensures transparency and auditability (Mislevy, 2004).

#### **5.1 Evidence as the Foundation**

The system relies on real-world evidence, including artifacts, observations, and feedback. Authentic assessment research shows that such evidence provides a more accurate representation of transferable skills than traditional testing (Pellegrino et al., 2001).

#### **5.2 Developmental Continuum**

Evidence is interpreted along a progression from emerging awareness to transformational influence, reflecting established theories of human development (Kegan, 1982; Dweck, 2006).

### **5.3 Integrated Capacities**

The system synthesizes skill signals into broader capacities such as agency, resilience, belonging, and trust, reflecting integrative models of human functioning (Bandura, 2001; Baumeister & Leary, 1995; Masten, 2014).

### **5.4 Multi-Source Verification**

A trust-weighted evidence model evaluates inputs based on source and type, strengthening inference validity through triangulation (AERA et al., 2014).

### **5.5 Confidence as a Separate Construct**

The system distinguishes between observed performance and certainty of interpretation, improving transparency and decision quality (Mislevy, 2004).

## **6. Incentive and Economic Model**

The system adopts a multi-sided platform approach in which learners, institutions, and employers each derive value. Employers benefit from improved hiring accuracy, institutions gain outcome visibility, and learners gain portable, verified records. This aligns with platform ecosystem theory, where value increases through network participation (Parker, Van Alstyne, & Choudary, 2016).

## **7. Calibration and Signal Normalization**

To ensure comparability across contexts, the system incorporates anchored developmental standards, rater adjustment models, and confidence weighting. Psychometric research demonstrates that such approaches improve fairness by accounting for variability in scoring conditions (Kolen & Brennan, 2014; Engelhard, 2013; Linacre, 1989).

## **8. Interoperability and System Integration**

The system is designed as an API-first architecture that integrates directly into existing platforms such as Applicant Tracking Systems and HR systems. Technology adoption research indicates that workflow integration is critical for sustained use (Davis, 1989). The system supports real-time data exchange and machine-readable skill representations aligned with global standards.

## **9. Human-AI Dispute Resolution**

A tiered dispute resolution model ensures scalability. Automated reconciliation handles most discrepancies, while peer review and expert adjudication address higher-stakes cases. Research shows that structured dispute systems reduce escalation and cost when clear processes are established (Lipsky, Seeber, & Fincher, 2003). Human-AI collaboration models further support efficient decision-making (Amershi et al., 2019).

## **10. Data Governance and Privacy**

The system adopts a learner-owned data model, ensuring individuals control access to their records. This aligns with global standards for verifiable credentials and emphasizes secure, interoperable data structures (W3C, 2022).

## **11. Implementation Model**

Implementation follows a phased approach: pilot programs, employer integration, and broader ecosystem expansion. This approach builds trust incrementally while enabling scalability through network effects.

## **12. Go-to-Market Strategy, Rater Usability, and Pilot Efficacy Framework**

### **Operationalizing Adoption, Reducing Cognitive Load, and Demonstrating Measurable ROI**

While the Human Capacity Intelligence System establishes a technically and psychometrically sound framework, its adoption depends on usability, targeted market entry, and measurable outcomes. This section defines how the system minimizes rater burden, identifies a strategic entry point, and establishes pilot metrics that demonstrate return on investment.

#### **12.1 Rater Burden and Usability Design**

A common failure point in assessment systems is excessive cognitive demand placed on evaluators. If structured rubrics require extensive interpretation or training, participation declines and data quality degrades. Research in human-computer interaction and assessment design demonstrates that systems must reduce cognitive load while preserving validity (Sweller, 1988; Nielsen, 1994).

The Skills Passport addresses this through three design principles:

##### **1. Progressive Disclosure of Complexity**

Rather than exposing full developmental frameworks, the system presents raters with:

- 2–3 observable indicators per skill
- Context-specific examples
- Pre-calibrated descriptors aligned to the user’s setting

This reduces decision fatigue while maintaining alignment with anchored standards.

## **2. Workflow-Embedded Input**

Evaluation occurs within existing processes:

- Internship evaluations
- Teacher grading workflows
- Performance reviews

Raters are not asked to complete new assessments; instead, existing inputs are structured into standardized formats. This aligns with research showing that embedding tools within workflows increases adoption and accuracy (Davis, 1989).

## **3. Just-in-Time Calibration**

Instead of requiring extensive upfront training, the system provides:

- Micro-calibration prompts
- Example comparisons (“This looks like Level 3 vs Level 4”)
- Immediate feedback on rating patterns

Studies on rater reliability show that ongoing calibration is more effective than one-time training sessions (Jonsson & Svingby, 2007).

## **4. Estimated Training Load**

- Initial onboarding: <30 minutes
- Ongoing interaction: <2 minutes per evaluation input

This ensures that participation remains feasible within real-world constraints.

# **12.2 Market Entry Strategy (The Wedge)**

A system designed for broad application must begin with a focused entry point where:

- The problem is acute
- The value is immediately measurable
- Stakeholders have incentive to adopt

Research on innovation diffusion shows that successful platforms begin in high-friction, high-value niches before expanding (Rogers, 2003).

## **Primary Wedge: Workforce-Aligned Education Pathways (Ages 16–24)**

The initial market focuses on:

- Community colleges
- Workforce development programs
- Career technical education (CTE) pathways

This segment is selected because it sits at the intersection of:

- Education systems (data-rich environments)
- Employers (urgent hiring needs)
- Learners (high need for signaling)

## **Why This Wedge Works**

### **1. Immediate Employer Pain**

Employers struggle to assess entry-level candidates without experience signals.

### **2. Policy Alignment**

Federal and state funding increasingly prioritizes workforce alignment and measurable outcomes.

### **3. Contained Ecosystems**

Programs already track cohorts, making pilot implementation feasible.

## **Expansion Path**

Phase 1: Workforce-aligned education programs

Phase 2: Employer hiring pipelines (entry-level roles)

Phase 3: Broader workforce credential ecosystem

This staged approach builds trust while expanding network effects.

## **12.3 Pilot Design and Proof of Efficacy**

For funders, conceptual validity must translate into measurable outcomes. Pilot programs are designed to generate evidence across three domains: hiring efficiency, retention, and performance alignment.

Research in workforce analytics emphasizes that improved job-person fit leads to measurable gains in productivity and retention (Boudreau & Cascio, 2017).

### **Pilot Structure**

- Duration: 6–12 months
- Participants: 100–500 learners
- Partners: 2–5 employers

- Context: Workforce-aligned training programs

## **Primary Outcome Metrics (KPIs)**

### **1. Hiring Efficiency**

- Time-to-hire reduction (%)
- Interview-to-offer ratio
- Screening accuracy (qualified candidates identified earlier)

### **2. Retention Outcomes**

- 90-day retention rates
- 6-month retention rates

Research indicates that improved fit significantly reduces early turnover, a major cost driver in workforce systems.

### **3. Performance Alignment**

- Supervisor-rated performance scores
- Skill application consistency
- Advancement or role progression

### **4. Signal Validity**

- Correlation between Skills Passport indicators and job performance
- Comparison with traditional credentials (e.g., GPA, degree)

### **5. Equity Indicators**

- Hiring rates across demographic groups
- Reduction in reliance on traditional proxies (e.g., institutional prestige)

Research suggests that skills-based approaches can reduce bias when properly implemented (National Research Council, 2012).

## **12.4 Success Benchmarks for Funding Decisions**

Pilot success is defined by:

- $\geq 20\%$  reduction in time-to-hire
- $\geq 15\%$  improvement in retention
- Positive correlation between verified capacities and performance
- Demonstrated employer adoption beyond pilot

These thresholds align with benchmarks used in workforce innovation evaluations.



## 12.5 Feedback and Iteration Loop

Pilot data feeds directly into system refinement through:

- Calibration updates
- Rubric adjustments
- Verification model tuning

This ensures continuous improvement and scalability.

## 12.6 Recruiter Experience and Signal Delivery

For the Skills Passport to influence hiring decisions, it must not only generate valid signals but present them in a format aligned to actual recruiter behavior. Recruiters often make first-pass decisions under severe time pressure, relying on fast heuristics rather than deep analysis (Rivera, 2012). If verified capacity data is too dense or difficult to interpret, recruiters may default back to legacy signals such as institutional prestige, GPA, or resume polish.

To address this, the Skills Passport uses a signal-compression model built around three layers of visibility.

### 1. The “3-Second View”

At the top layer, recruiters see:

- Top verified capacities
- Evidence strength framed in plain language
- Context tags indicating where the capacity was demonstrated

This creates a rapid-read signal that can function alongside existing screening artifacts.

### 2. The “10-Second View”

With a click or hover, recruiters can access:

- A concise evidence summary
- Developmental level
- Verification density across sources

This gives decision-makers just enough depth to confirm credibility without forcing a deep dive.

### 3. The “Deep Dive”

For candidates of interest, recruiters can view:

- Evidence timeline
- Source breakdown
- Full developmental progression

This layered design ensures that the Skills Passport augments recruiter workflow rather than competing with it.

## 12.7 Anchor Payor Strategy

A critical product-market-fit challenge is defining who pays during initial adoption. In the 16–24 education-to-workforce wedge, institutional budgets are constrained and employer-funded models can be perceived as a hiring tax. For this reason, the most viable early-stage anchor payor is the public workforce and economic development ecosystem.

Initial deployment is best positioned for:

- State workforce agencies
- Regional workforce development boards
- Public-private workforce grants

This strategy aligns the Skills Passport with existing investments in labor-market efficiency, workforce readiness, and equitable access to opportunity. By positioning the system as infrastructure that improves the return on workforce development spending, early deployment can occur without shifting immediate cost to learners, community colleges, or employers (Autor, 2014).

A phased transition model can then follow:

- Phase 1: Publicly subsidized access
- Phase 2: Employer co-investment
- Phase 3: Institutional subscription

## 12.8 Confidence Signal Safeguards

Confidence is one of the system's most important constructs, but it also presents a risk if misinterpreted. In a high-volume screening environment, a recruiter might read a strong capacity signal paired with low confidence and treat it as grounds for rejection. This risk is particularly acute for nontraditional learners who may have fewer formal verification sources despite strong underlying capacity.

To mitigate this, the Skills Passport does not frame confidence as a deficit indicator. Instead, it is presented as **evidence strength**, using language such as:

- Emerging evidence
- Established evidence
- Verified evidence

This framing reduces the likelihood that uncertainty will be interpreted as weakness.

In addition, confidence is contextualized relative to opportunity exposure. Learners from under-resourced settings may have fewer opportunities for formal validation. The system therefore interprets evidence density with attention to access, rather than treating thin evidence as inherently low capability (Darling-Hammond, 2010).

Finally, recruiter-facing views prioritize verified capacities first and present evidence strength only as secondary context. This ensures that capability remains the primary signal in first-pass review.

## 12.9 Zero-Integration Employer Adoption

Although the Skills Passport is designed for API-first interoperability, early employer adoption cannot depend solely on technical integration. Many organizations, especially larger enterprises, operate within legacy systems and long IT procurement cycles. If adoption requires immediate integration work, early traction will be slowed.

To solve this, the Skills Passport supports a zero-integration pathway through a **Verified Portable Profile** model. Each learner can generate:

- A secure digital profile link
- A downloadable verified PDF
- A machine-verifiable QR code linking to a live authenticated record

This allows employers to use Skills Passport data immediately within existing hiring processes, without API work. Recruiters can review a web view, attach a verified PDF to candidate files, or share internally with no technical lift.

As value is demonstrated, organizations can progress through a staged integration path:

- Stage 1: Manual review through link or PDF
- Stage 2: Lightweight uploads or attachments
- Stage 3: Full API integration

This approach reduces adoption friction and aligns with established enterprise adoption patterns (Rogers, 2003).

## 12.10 Dispute Ratio, Adjudication Cost, and Unit Economics

A final operational concern is whether dispute resolution will become too expensive at scale. Human expert review is costly, so the economics of the system depend not on eliminating disputes but on keeping the expert-adjudication rate low.

The Skills Passport uses a tiered dispute model:

- Tier 1: Automated reconciliation
- Tier 2: Structured secondary review

- Tier 3: Expert adjudication

Under this design, the majority of disputes are expected to resolve before reaching expert review. A reasonable projected dispute-to-verification structure is:

- Total disputes: 3–7% of evaluations
- Tier 1 resolutions: ~80%
- Tier 2 resolutions: ~15%
- Tier 3 expert adjudication: ~5% of disputes

This results in expert intervention for only a very small fraction of total evaluations, helping maintain sustainable per-user costs.

The economic value of the system is strengthened by the downstream benefits of improved signal quality. Better screening and fit can reduce time-to-hire, improve retention, and lower mismatch costs. Because turnover can impose substantial financial burdens on employers, even modest improvements in hiring accuracy can create meaningful return on investment (Boudreau & Cascio, 2017).

## 13. Conclusion

The central challenge facing education and workforce systems is not the absence of human capacity development, but the absence of systems capable of capturing, verifying, and translating that development into meaningful signals.

By integrating evidence, development, verification, and interoperability, the Skills Passport provides a viable pathway toward more equitable, efficient, and aligned systems of learning and opportunity.

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